LIN Communication Usage

Marek Ďurďa, Pavel Musil, Ondřej Ondříšek

This project introduces the LIN communication protocol and provides practical hardware and software design guidelines using the MSPM0C1104 microcontroller and theTLIN1039 LIN transceiver

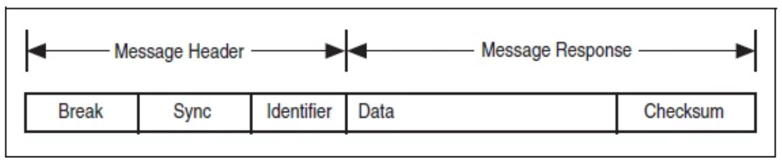

Figure 1: LIN frame structure

TLIN1039 LIN transceiver

- Merges UART TX/RX onto a single LIN bus pin
- Translates MCU logic to automotive LIN voltage levels
- Includes bus-stuck-low protection (prevents a node from holding LIN low)
- Implements ESD protection and slew-rate control for EMC compliance
- Provides short-circuit protection on the LIN pin
- Supports low-power modes and fault detection

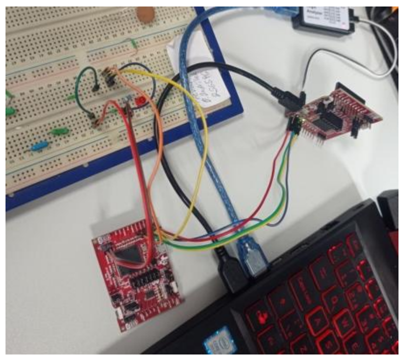

Figure 2: MCUs wire-up

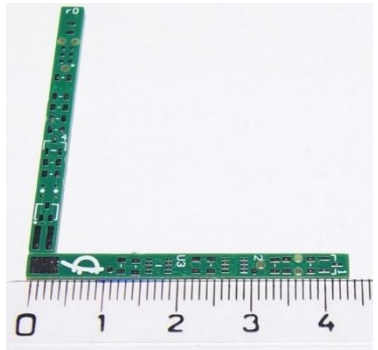

Figure 3: ground temperature sensor PCB

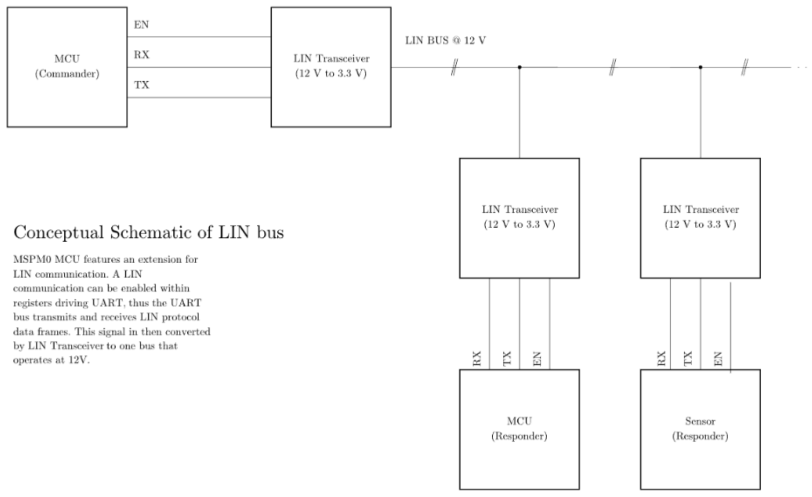

Figure 4: LIN bus schematic

Demonstration

The LIN bus was selected as the communication protocol for collecting data from ground temperature sensors as part of final thesis “System for temperature control and security surveillance of turtle breeding”. The choice was motivated primarily by the bus’s reliable range and simplicity. Since the temperature sensor will be in a small metallic tube, the use of a single data line is advantageous, minimizing wiring complexity and ensuring robust signal transmission.

Sensor schematic description

MSPM0C1103 microcontroller was selected for its enhanced UART capabilities, which make LIN implementation easier. An analog temperature sensor was chosen for SW simplicity. For programming and debugging, a Tag-Connect interface is implemented, providing a compact, solder-free connection directly to the PCB. A key challenge in the design is that the SWD programming pins are multiplexed with other functional signals, which is an inherent trade-off of the small package footprint (SOT-8). For circuit protection, ESD and Schottky diodes were used.

Software project description

After initialization, the commander waits for a button event (simulating RTC) to trigger the communication cycle. Once activated, the commander sequentially polls all defined PIDs and collects the responses from the responder MCU.

The implementation uses a flag-driven approach: on button press, an event flag is set, and the main loop checks this flag to perform the corresponding action. For demonstration purposes, the responder’s code reacts to incoming commands by toggling its built-in LED, providing a simple visual confirmation of successful data exchange.

Figure 5: Flow chart of LIN commander

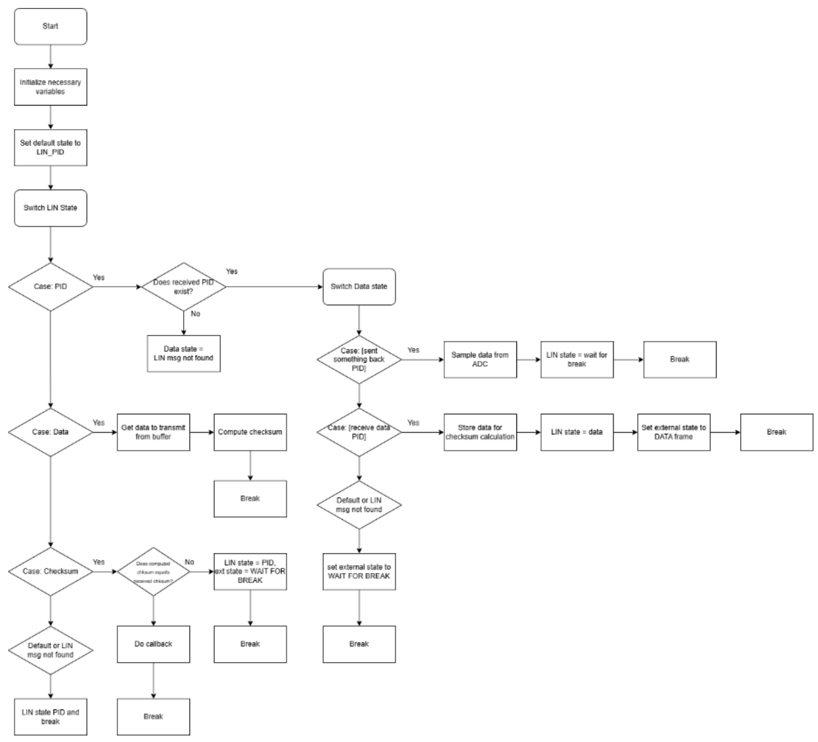

Figure 6: Flow chart of LIN responder

Checksum

To ensure reliable transmission, LIN communication employs a checksum calculated as the sum of all transmitted bytes and the PID modulo 256. The checksum format can be chosen arbitrarily, provided that all nodes use the same calculation method. It may be positive, meaning the raw checksum value is sent, or negative, where the final checksum is subtracted from 0xFF. Upon receiving a message, the responder immediately computes its own checksum and compares it with the received value, thereby verifying the integrity of the transmission.

Conclusion

We designed and implemented LIN communication, verified its operation on the LIN bus, and prepared the system for deployment in a real-world application.

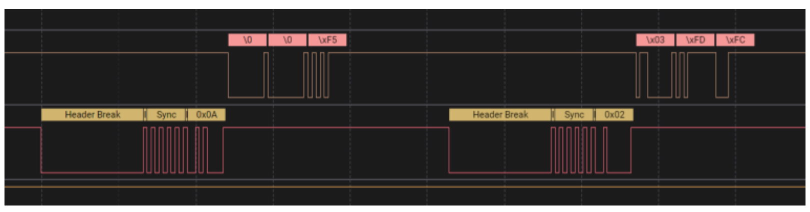

Figure 7: Demonstration of polling

References

- The ground temperature LIN sensor is part of the semestral thesis: MUSIL, Pavel. System for temperature control and security surveillance of turtle breeding. Semestral Thesis. Roman ŠOTNER (supervisor). Brno: Brno University of Technology, Faculty of Electrical Engineering and Communication, 2026.
- The development of the ground temperature LIN sensor was consulted with engineers from egmenergo, in particular with Ing. Ivo Strašil, who is a technical consultant for the thesis.
- MSPM0C1104 Technical Reference Manual (SLAU893C)
- TLIN1039 LIN Transceiver Datasheet
- LIN 2.0/2.1/2.2A Protocol Specifications